

EXERCISE 2.2

Prove the following:

1. $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$
2. $3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1\right]$
3. $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \frac{1}{2}$
4. $2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

Write the following functions in the simplest form:

- 5 $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right), \quad x \neq 0$
- 6 $\tan^{-1} \left(\frac{1}{\sqrt{x^2-1}} \right), \quad |x| > 1$
- 7 $\tan^{-1} \left(\sqrt{\frac{1-\cos(x)}{1+\cos(x)}} \right), \quad 0 < x < \pi$
- 8 $\tan^{-1} \left(\frac{\cos(x)-\sin(x)}{\cos(x)+\sin(x)} \right), \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$
- 9 $\tan^{-1} \left(\frac{x}{\sqrt{a^2-x^2}} \right), \quad |x| < a$
- 10 $\tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right), \quad a > 0, \quad -\frac{a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$

Find the values of each of the following:

- 11 $\tan^{-1} \left[2 \cos \left(2 \sin^{-1} \left(\frac{1}{2} \right) \right) \right]$
- 12 $\sin \left(\sin^{-1} \left(\frac{1}{5} \right) + \cos^{-1}(x) \right) = 1, \text{ find } x$
- 13 $\tan \left(\frac{1}{2} \right) \left[\sin^{-1} \left(\frac{2x}{1+x^2} \right) + \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right], \quad |x| < 1, \quad y > 0, \quad xy < 1$
- 14 $\sin \left(\sin^{-1} \left(\frac{1}{5} \right) + \cos^{-1}(x) \right) = 1, \text{ find } x$
- 15 $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x-2} \right) = \frac{\pi}{4}, \text{ find } x$

Find the values of each of the expressions in Exercises 16 to 18.

- 16 $\sin^{-1} \left(\sin \frac{2\pi}{3} \right)$
- 17 $\tan^{-1} \tan \frac{3\pi}{4}$
- 18 $\tan \left(\sin^{-1} \frac{3}{5} + \cos^{-1} \frac{7}{25} \right)$
- 19 $\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$ is equal to:

- (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

- 20 $\sin \left(\frac{3\pi}{2} - \sin^{-1} \frac{1}{2} \right)$ is equal to:

- (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) π

- 21 $\tan^{-1} \sqrt{3} - \cos^{-1}(\sqrt{3})$ is equal to:

- (A) π (B) $\frac{\pi}{2}$ (C) 0 (D) $2\sqrt{3}$